STACK:

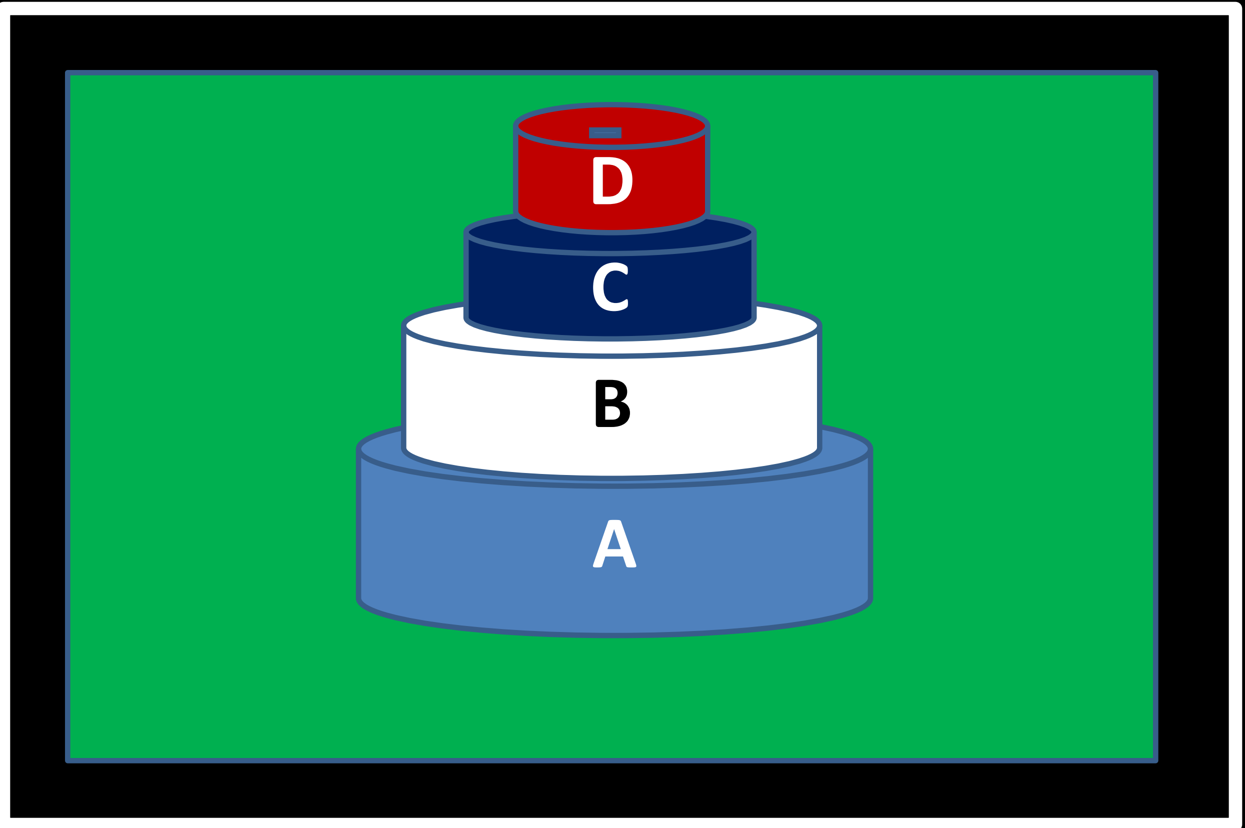
A pole is placed on a plane.

Disk A is placed in the pole.

Disk B is placed on disk A.

Disk C is placed on disk B.

Disk D is placed on disk C.



This type of operation is called **push** instead of insert.

Where to push? ***At TOP***

Is it possible to another disk E?

If there is no space to push in STACK, the situation is called overflow.

STACK:

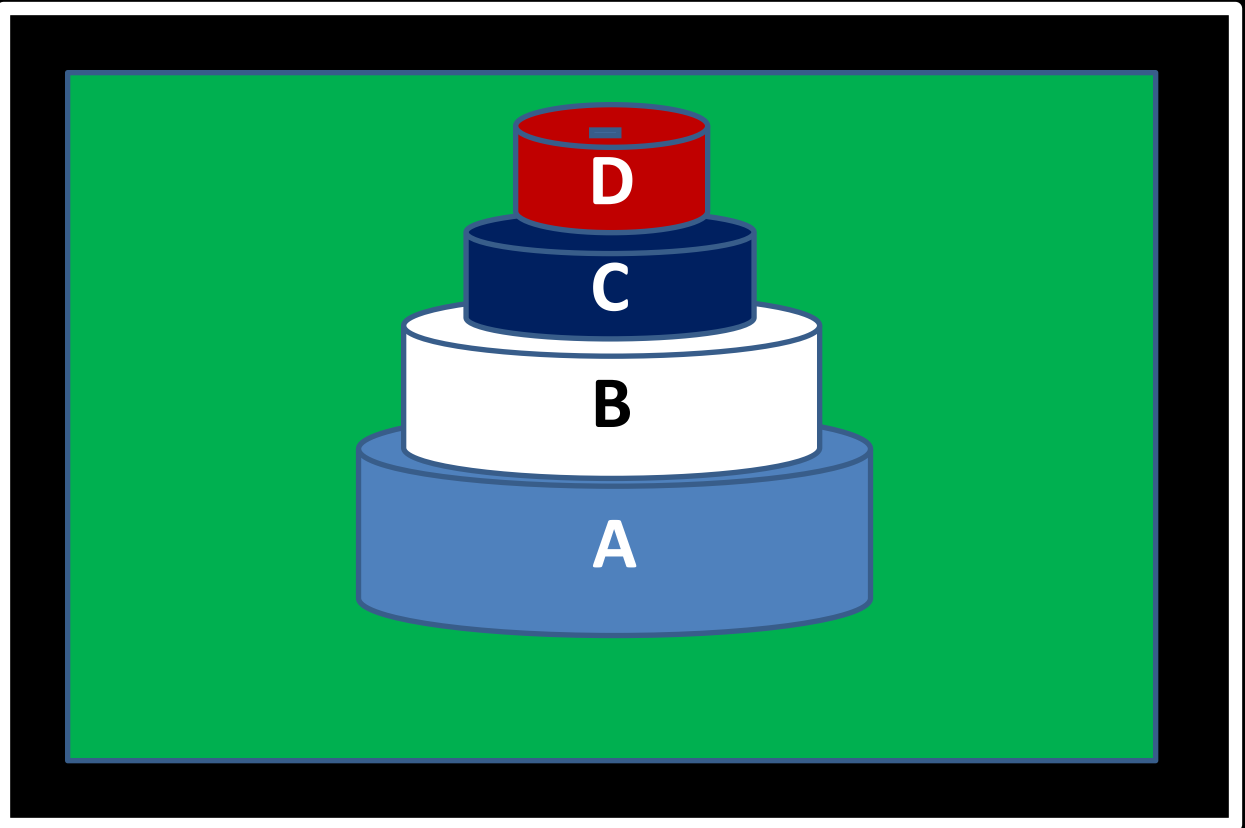
A pole is placed on a plane.

Disk A is placed in the pole.

Disk B is placed on disk A.

Disk C is placed on disk B.

Disk D is placed on disk C.



This type of operation is called **push** instead of insert.

Where to push? ***At TOP***

Now remove disk D.

Now remove disk C.

Now remove disk B.

Now remove disk A.

Removing/deleting process is known as POP.

Is it possible to POP more?

Ans: No, because the pole is empty. When the STACK is empty but one tries to perform POP, the situation is called underflow.

STACK:

Elements are
AAA, BBB, CCC, DDD, EEE, FFF.

Sequentially pushed in an array of STACK.

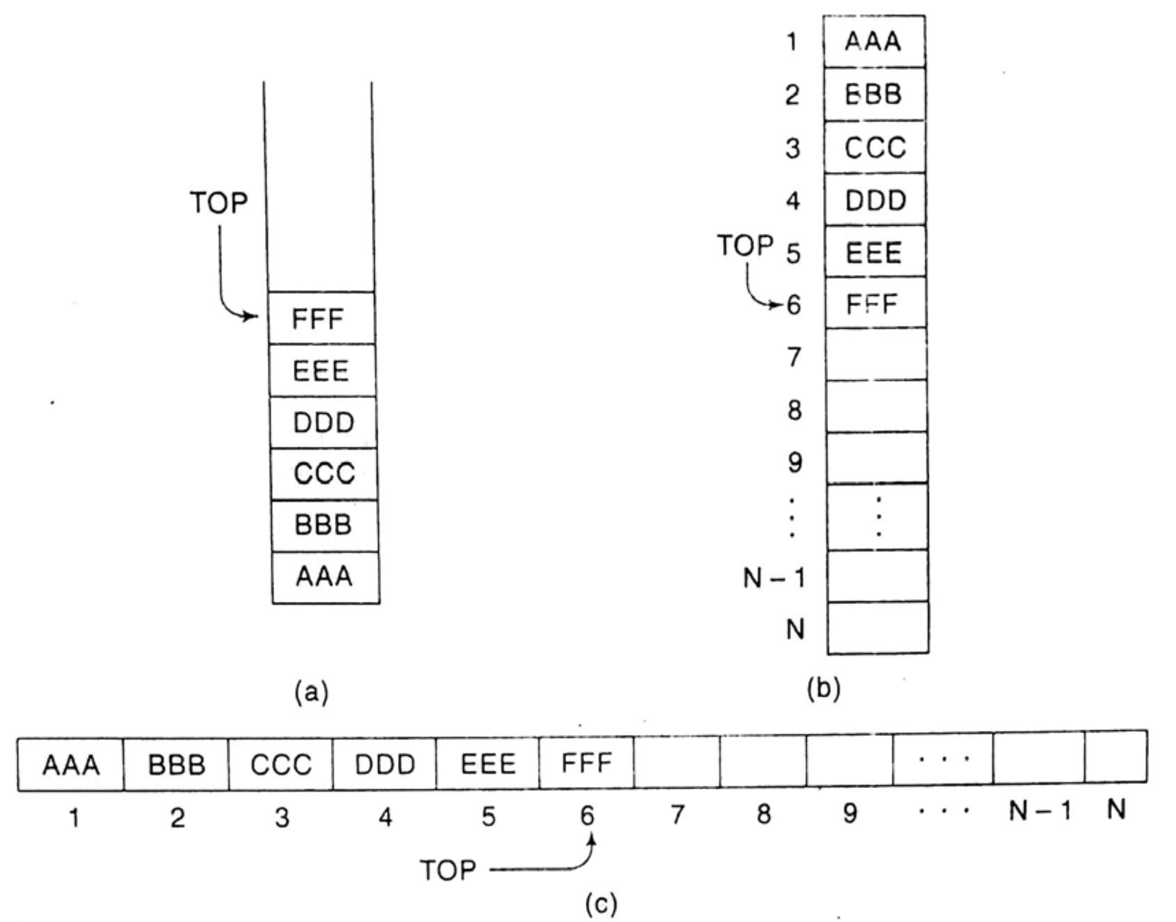


Fig. 6.3 *Diagrams of Stacks*

STACK:

Elements are A, B, C.

Sequentially pushed in an array of STACK.

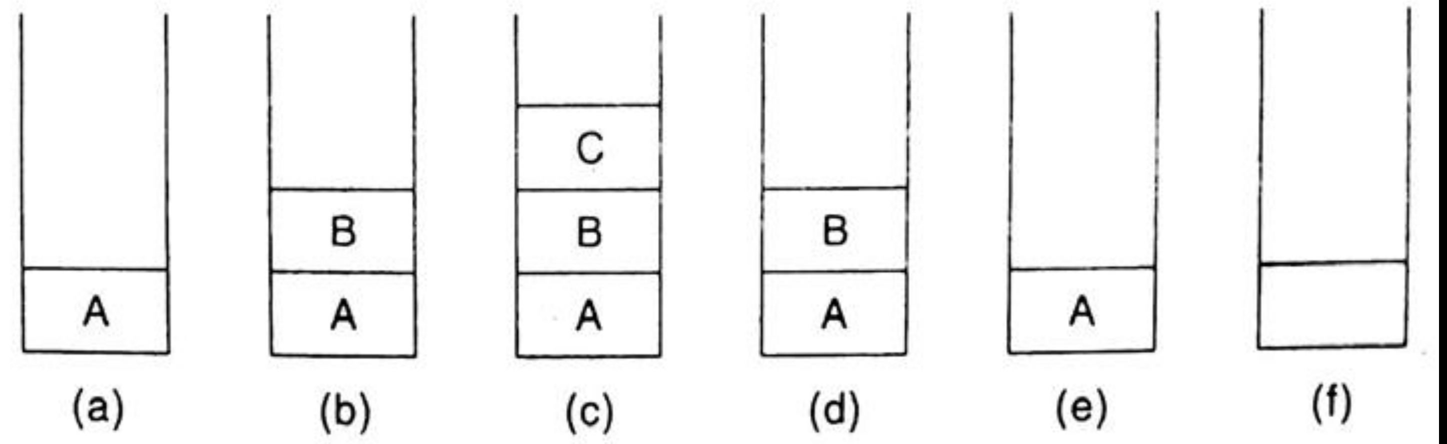


Fig. 6.4

STACK: Array representation

Elements are XXX, YYY, ZZZ.

Sequentially pushed in an array of STACK.

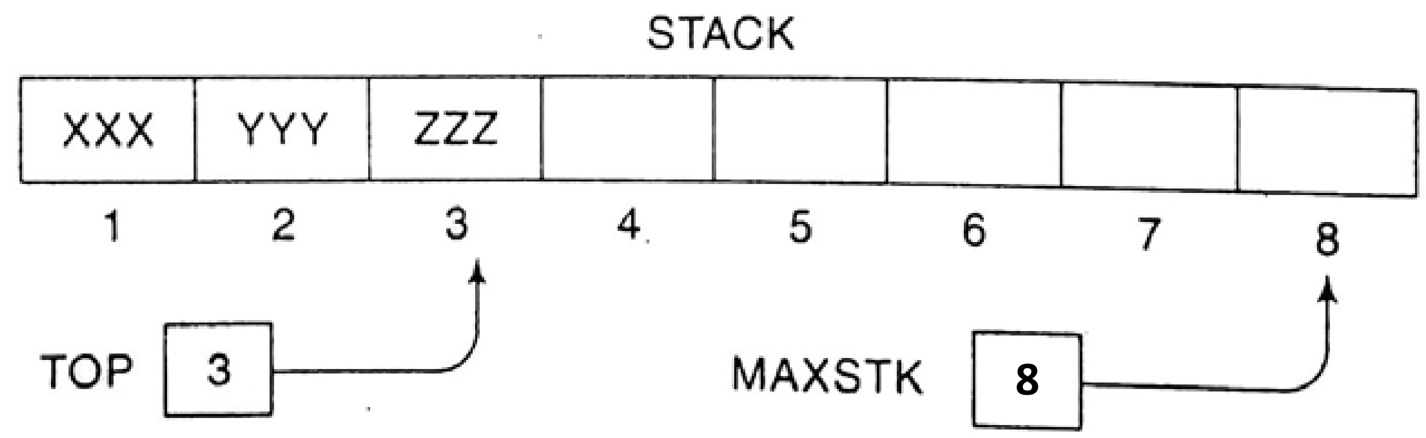


Fig. 6.5

STACK: Array representation

Let, ITEM="WWW"
Here,
MAXSTK=11
TOP=3

Procedure 6.1:

PUSH(STACK, TOP, MAXSTK, ITEM)
This procedure pushes an ITEM onto a stack.

1. [Stack already filled?]
   If TOP = MAXSTK, then: Print: OVERFLOW, and Return.
2. Set TOP := TOP + 1. [Increases TOP by 1.]
3. Set STACK[TOP] := ITEM. [Inserts ITEM in new TOP position.]
4. Return.

Step 2. TOP=TOP+1=3+1=4
Step 3. STACK[TOP]=STACK[4]=WWW

Array of STACK:

AAA	BBB	CCC								
1	2	3	4	5	6	7	8	9	10	11

STACK: Array representation

Let, ITEM="WWW"
Here,
MAXSTK=11
TOP=3

Procedure 6.1:

PUSH(STACK, TOP, MAXSTK, ITEM)
This procedure pushes an ITEM onto a stack.

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Step 2. TOP=TOP+1=3+1=4
Step 3. STACK[TOP]=STACK[4]=WWW

Array of STACK:

AAA	BBB	CCC	WWW							
1	2	3	4	5	6	7	8	9	10	11

STACK: Array representation

Here,
MAXSTK=11
TOP=4

Procedure 6.1:

POP(STACK, TOP, ITEM)
This procedure deletes the top element of STACK and assigns it to the variable ITEM.

1. [Stack has an item to be removed?]
   If TOP = 0, then: Print: UNDERFLOW, and Return.
2. Set ITEM := STACK[TOP]. [Assigns TOP element to ITEM.]
3. Set TOP := TOP – 1. [Decreases TOP by 1.]
4. Return.

Step 2. ITEM=STACK[TOP]=STACK[4]=WWW
Step 3. TOP=TOP-1=3

Array of STACK:

AAA	BBB	CCC	WWW							
1	2	3	4	5	6	7	8	9	10	11

STACK: Performing Arithmetic Operation

Notations:

- Infix
- Polish (Prefix)
- Reverse Polish (Postfix)

STACK: Performing Arithmetic Operation

Notations:

- **Infix**
- Polish (Prefix)
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(A+B)*C

STACK: Performing Arithmetic Operation

Notations:

- Infix
- **Polish (Prefix)**
- Reverse Polish (Postfix)

Operator symbol is placed before its two operand:

In Infix: (A+B)*C
In Polish: *+ABC

$$(A+B)/(C-D) = [+AB]/[-CD] = /+AB-CD$$

Infix: 12/(7-3)+2*(1+5)
In Polish: 12/[-, 7, 3]+2*[+, 1, 5]
: [/12,-,7,3]+[*, 2, +, 1, 5]
:+, /, 12, -, 7, 3, *, 2, +, 1, 5

STACK: Performing Arithmetic Operation

Notations:

- Infix
- **Polish (Prefix)**
- Reverse Polish (Postfix)

Operator symbol is placed before its two operand:

In Infix: (A+B)*C
In Polish: *+ABC

P: +, /, 12, -, 7, 3, *, 2, +, 1, 5
P: +, /, 12, 4, *, 2, +, 1, 5
P: +, 3, *, 2, +, 1, 5
P: +, 3, *, 2, 6
P: +, 3, 12
P: 15

$$12/(7 - 3) + 2 * (1 + 5)$$

STACK: Performing Arithmetic Operation

Notations:

- Infix
- Polish (Prefix)
- **Reverse Polish (Postfix)**

Operator symbol is placed after its two operand:

In Infix: (A+B)*C

In Postfix: AB+C*

P= 5, $\boxed{6,\quad 2,\quad +,}$ *, 12, 4, /, -

P= $\boxed{5,\quad 8,\quad *,}$ 12, 4, /, -

P= 40, $\boxed{12,\quad 4,\quad /,}$ -

P= $\boxed{40,\quad 3,\quad -}$

P= 37

STACK: Performing Arithmetic Operation

Recursion:
Calling itself.

Factorial

Fibonacci Sequence

Tower of Hanoi: